

KHOI NGUYEN

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EDUCATION

University of North Carolina at Chapel Hill

May 2026

B.A Geography | Minor in GIS

Major & Minor GPA: 3.3

TECHNICAL SKILLS

Enterprise GIS, Servers & Databases: ArcGIS Pro, ArcGIS Enterprise, Portal, Server, Data Store, ArcGIS Online, Hub; PostgreSQL/PostGIS, GeoServer, QGIS, pgAdmin, Docker; hosted/referenced services, WMS/WFS/WMTS, metadata, sharing, public data workflows, spatial indexes, materialized views, and backup/restore

Web GIS, BI & Spatial Intelligence: ArcGIS Maps SDK, MapView, Next.js, TypeScript; Power BI-ready exports, KPI dashboards, parcel intelligence, development dashboards, scenario planning and infrastructure readiness.

GIS Automation & Spatial Data Management: Python, ArcPy, GeoPandas, ModelBuilder; geodatabases, shapefiles, batch geoprocessing, QA/QC, schema/projection validation, raster/vector processing, and REST service review

RELATED EXPERIENCE

GIS Analyst Intern — Cabarrus County, NC

March 2026 – Present

- Independently redesigned Cabarrus County's public Open Data / GIS Hub, improving navigation, discoverability, and access to county GIS resources.
- Managed public-facing GIS items through ArcGIS Hub, ArcGIS Online, and ArcGIS Enterprise/Portal, including hosted layers, web maps, metadata, sharing settings, and access links.
- Gained hands-on experience with ArcGIS Enterprise setup and configuration workflows, including Portal, ArcGIS Server, ArcGIS Data Store, Web Adaptor concepts, database connections, registered data sources, and service publishing on new machines.
- Used Python scripts to automate GIS Hub item review, tagging, category assignment, metadata cleanup, and access-link validation for public-facing ArcGIS Online and Enterprise/Portal content.
- Used Python scripts to automate GIS Hub and Enterprise item review, including tagging, category assignment, metadata cleanup, broken-link checks, and public-sharing validation for county GIS resources.

Cabarrus FutureScape — County Digital Twin & Growth Intelligence Platform

Project website: <https://cabarrus-future-scape.vercel.app/>

- Designed a county-scale planning intelligence prototype for Cabarrus County to support parcel screening, development review, growth-pressure analysis, and long-range land use and infrastructure planning decisions.
- Built an interactive Web GIS dashboard using Next.js, TypeScript, ArcGIS Maps SDK, SceneView, operational layers, parcel selection, planning summaries, and dashboard components.
- Developed decision-support workflows for reviewing parcel context, development activity, constraints, infrastructure readiness, school-capacity context, and growth hotspots.
- Built CFS Economics, a parcel-based economic intelligence module that screens tax-base opportunity, underbuilt parcels, redevelopment signals, fiscal/service-burden indicators, and data-confidence issues across countywide property data.
- Designed Power BI-ready exports, scenario outputs, KPI summaries, opportunity-class visuals, underbuilt parcel watchlists, value-per-acre comparisons, and printable decision snapshots to support planning and economic development analysis.

Enterprise GIS Systems Lab — PostGIS, GeoServer, QGIS & Docker

- Built a local enterprise GIS lab with Docker, PostgreSQL/PostGIS, GeoServer, pgAdmin, and QGIS to practice database-backed GIS service publishing.
- Designed spatial database schemas for core and analytics layers, including parcel, permit, floodplain, and development-pressure datasets with geometry columns, spatial indexes, and primary keys.
- Practiced GIS recovery workflows using pg_dump, pg_restore, Docker volumes, and GeoServer data-directory backup/restore.
- Published PostGIS layers through GeoServer as WMS, WFS, and layer-group services, including WFS-ready feature layers with stable IDs for QGIS attribute/query access.

ADDITIONAL INFORMATION

Languages: English (Fluent), Vietnamese (Fluent)